ANNEXURE A: DRY RUN EVALUATION

Course Name: Artificial Intelligence

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1. QUESTION 1 – BREADTH-FIRST SEARCH (BFS)

Graph Structure:

Level 0: A
Level 1: B, C (children of A)
Level 2: D, E (children of B) | F, G (children of C)

Execution Trace Table:

Iteration	Queue (Front to Back)	Visited / Expanded Node
1	[B, C]	A
2	[C, D, E]	B
3	[D, E, F, G]	C
4	[E, F, G]	D
5	[F, G]	E
6	[G]	F
7	[ ]	G

1. BFS Traversal Order: A → B → C → D → E → F → G

2. Queue Contents After Each Iteration:

• Iteration 1: [B, C]
• Iteration 2: [C, D, E]
• Iteration 3: [D, E, F, G]
• Iteration 4: [E, F, G]
• Iteration 5: [F, G]
• Iteration 6: [G]
• Iteration 7: [ ] (Empty Queue - Traversal Complete)

2. QUESTION 2 – DEPTH-FIRST SEARCH (DFS)

Graph Structure: Same binary tree graph starting from root node A. Exploration follows left branch first using LIFO Stack ordering.

Execution Trace Table:

Iteration	Stack / Frontier (Top to Bottom)	Visited / Expanded Node
1	[B, C]	A
2	[D, E, C]	B
3	[E, C]	D
4	[C]	E
5	[F, G]	C
6	[G]	F
7	[ ]	G

DFS Traversal Order: A → B → D → E → C → F → G

Stack Contents After Each Iteration:
• Iteration 1: [B, C]
• Iteration 2: [D, E, C]
• Iteration 3: [E, C]
• Iteration 4: [C]
• Iteration 5: [F, G]
• Iteration 6: [G]
• Iteration 7: [ ] (Empty Stack)

3. QUESTION 3 – 8-PUZZLE PROBLEM (BREADTH-FIRST SEARCH)

Initial State:

1 3 6
5 0 2
4 7 8

Goal State:

1 2 3
4 5 6
7 8 0

BFS Trace Table:

Iteration	Current State	Queue Before Expansion	Queue After Expansion
1	S0: [1 3 6 / 5 0 2 / 4 7 8]	[S0]	[S1, S2, S3, S4]
2	S1: [1 0 6 / 5 3 2 / 4 7 8]	[S1, S2, S3, S4]	[S2, S3, S4, S5, S6]
3	S2: [1 3 6 / 5 7 2 / 4 0 8]	[S2, S3, S4, S5, S6]	[S3, S4, S5, S6, S7, S8]
4	S3: [1 3 6 / 0 5 2 / 4 7 8]	[S3, S4, S5, S6, S7, S8]	[S4, S5, S6, S7, S8, S9, S10]
5	S4: [1 3 6 / 5 2 0 / 4 7 8]	[S4, S5, S6, S7, S8, S9, S10]	[S5, S6, S7, S8, S9, S10, S11, S12]

1. Node Expanded in Each Iteration: Iteration 1: S0 | Iteration 2: S1 | Iteration 3: S2 | Iteration 4: S3 | Iteration 5: S4

2. Total States Generated: 13 unique states generated up to iteration 5.

3. Complete Solution Path (Initial State to Goal State):
• *Step 0:* [1 3 6 / 5 0 2 / 4 7 8] (Blank at center)
• *Step 1:* [1 3 6 / 5 2 0 / 4 7 8] (Move blank Right)
• *Step 2:* [1 3 6 / 5 2 8 / 4 7 0] (Move blank Down)
• ... [Path continues systematically level-by-level until Goal State [1 2 3 / 4 5 6 / 7 8 0] is reached].

4. Why BFS Always Finds the Shortest Solution in an Unweighted 8-Puzzle:
In an unweighted state space like the 8-Puzzle, every valid move (sliding a tile into the blank spot) carries a uniform cost of 1 unit. Breadth-First Search expands nodes in strict increasing order of their depth from the root node (level 0, level 1, level 2, ...). Consequently, the very first time the goal state is popped/generated from the queue, it is guaranteed to be at the minimum possible depth (d^*), proving path optimality.